Decimals Worksheet 2 – 1st ESO

Topic: Decimals: Ordering, Rounding and Conversion
Level: 1st Year Secondary Education (1st ESO)

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1. Theoretical Summary

Decimals allow us to express parts of a whole in a clear way. When working with measurements like radio frequencies, decimals let us be very precise. For example, 88.75 MHz means 88 whole units and 75 hundredths.

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2. Exercises

1. Arrange these frequencies from lowest to highest:
   90.1 MHz, 87.5 MHz, 89.3 MHz, 88.75 MHz
2. Round the following frequencies to one decimal place:
   - 88.76 MHz
   - 87.45 MHz
   - 89.31 MHz
3. Convert these frequencies from MHz to Hz:
   - 88.7 MHz
   - 90.1 MHz
4. Convert these frequencies from MHz to kHz:
   - 88.75 MHz
   - 89.3 MHz
5. Write the decimal number 87.556 rounded to:
   - One decimal place
   - Two decimal places
   - Three decimal places

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3. Competency Problems

1. Your radio tunes to 88.76 MHz but the screen shows 88.8 MHz after rounding. What is the difference?
2. If you want to save stations with frequencies rounded to the nearest 0.1 MHz, what frequency will 87.45 MHz be saved as?
3. Convert 90.1 MHz into Hz and explain why it is important to use such large numbers when programming radio modules.

4. The radio can tune frequencies with steps of 0.05 MHz. How many steps are there between 87.5 MHz and 90.0 MHz?
5. Explain why decimals are important when storing and retrieving frequencies on your FM radio.

Solutions

Exercises:

1. Sorted frequencies: 87.5 MHz, 88.75 MHz, 89.3 MHz, 90.1 MHz
2. Rounded to one decimal place:
   - 88.76 → 88.8 MHz
   - 87.45 → 87.5 MHz
   - 89.31 → 89.3 MHz
3. MHz to Hz:
   - 88.7 MHz = 88,700,000 Hz
   - 90.1 MHz = 90,100,000 Hz
4. MHz to kHz:
   - 88.75 MHz = 88,750 kHz
   - 89.3 MHz = 89,300 kHz
5. Rounding 87.556:
   - One decimal: 87.6
   - Two decimals: 87.56
   - Three decimals: 87.556

Competency Problems:

1. Difference = 88.8 - 88.76 = 0.04 MHz
2. 87.45 MHz rounded to nearest 0.1 MHz = 87.5 MHz
3. 90.1 MHz = 90,100,000 Hz; large numbers ensure accuracy in tuning and signal processing.
4. Number of steps: (90.0 - 87.5) / 0.05 = 50 steps
5. Decimals allow precise tuning, saving, and differentiating between stations close in frequency.